

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
#Loading necessary packages
library(tidyverse)
library(lubridate)
library(here)

#Checking working directory
getwd()
```

```
## [1] "/home/guest/ciraortizenviron872"
```

```
here()
```

```
## [1] "/home/guest/ciraortizenviron872"
```

```
#Uploading Neonics dataset
```

```
Neonics<-read.csv(
```

```
  file=here('./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv'),  
  stringsAsFactors = TRUE)
```

```
#Neonics
```

```
#Uploading Niwot Ridge NEON dataset for litter and woody debris
```

```
Litter<-read.csv(
```

```
  file=here('./Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv'),  
  stringsAsFactors = TRUE)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: The use of neonicotinoids can have adverse effects on insects that then impact the ecosystem they are part of. This could lead to negative impacts on pollination, water health, and soil health, as insects all play important roles in regulating these elements of ecosystems. Additionally, depending on how these neoincotinoids affect insects, they can also have trophic cascade effects at other levels within an ecosystem.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: These areas filled with forest litter and woody debris play important roles in the health of an ecosystem. From nutrient cycling, to improving soil health, and even being a habitat for many insects and fungi. Studying these sites can shed light on external stressors and processes that are impacting these sites and affecting overall ecosystem health.

4. How is litter and woody debris sampled as part of the NEON network? Read the [NEON_Litterfall_UserGuide.pdf](#) document to learn more. List three pieces of salient information about the sampling methods here:

Answer: The litter and woody debris was collected from elevated and ground traps. The mass data had several functional groups such as leaves, needles, seeds, and twigs/branches. Each functional group had a corresponding weight. 1. Ground traps were only sampled once per year. 2. Depending on the type of site (evergreen or deciduous forests) the sites were samples once very 2 weeks or once ever 1-2 months. 3. Sites that experienced severe winter conditions or had deciduous vegetation were sometimes not sampled for up to 6 months.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
#Finding dimensions of Litter dataset - you can also see this in the environment pane under the data  
#Dimensions of Litter dataset are 188 obs x 19 variables  
dim(Litter)
```

```
## [1] 188 19
```

```
#Finding dimensions of Neonics dataset  
#Dimensions of Neonics dataset are 4623 obs x 30 variables  
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
#Using summary to get view of effect column  
summary(Neonics$Effect)
```

```
##      Accumulation      Avoidance      Behavior      Biochemistry  
##           12           102           360           11  
##      Cell(s)      Development      Enzyme(s)      Feeding behavior  
##           9           136           62           255  
##      Genetics      Growth      Histology      Hormone(s)  
##          82           38           5           1  
##      Immunological      Intoxication      Morphology      Mortality  
##          16           12           22           1493  
##      Physiology      Population      Reproduction  
##           7           1803           197
```

```
#sorting the summary effect column within a table  
sort(table(Neonics$Effect))
```

```
##  
##      Hormone(s)      Histology      Physiology      Cell(s)  
##           1           5           7           9  
##      Biochemistry      Accumulation      Intoxication      Immunological  
##          11           12           12           16  
##      Morphology      Growth      Enzyme(s)      Genetics  
##          22           38           62           82  
##      Avoidance      Development      Reproduction      Feeding behavior  
##          102           136           197           255  
##      Behavior      Mortality      Population  
##          360           1493           1803
```

#> Answer: These effects can show provide insight as to how the neoincotinoids are affecting biological and physical characteristics of insects. Consequently, the effects can be analyzed to understand how they're impacting the greater ecosystem.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
#getting summary of species
```

```
summary(Neonics$Species.Common.Name)
```

```
##              Honey Bee              Parasitic Wasp
##              667              285
##      Buff Tailed Bumblebee      Carniolan Honey Bee
##              183              152
##              Bumble Bee              Italian Honeybee
##              140              113
##      Japanese Beetle              Asian Lady Beetle
##              94              76
##      Euonymus Scale              Wireworm
##              75              69
##      European Dark Bee              Minute Pirate Bug
##              66              62
##      Asian Citrus Psyllid              Parastic Wasp
##              60              58
##      Colorado Potato Beetle              Parasitoid Wasp
##              57              51
##      Erythrina Gall Wasp              Beetle Order
##              49              47
##      Snout Beetle Family, Weevil              Sevenspotted Lady Beetle
##              47              46
##      True Bug Order              Buff-tailed Bumblebee
##              45              39
##      Aphid Family              Cabbage Looper
##              38              38
##      Sweetpotato Whitefly              Braconid Wasp
##              37              33
##      Cotton Aphid              Predatory Mite
##              33              33
##      Ladybird Beetle Family              Parasitoid
##              30              30
##      Scarab Beetle              Spring Tiphia
##              29              29
##      Thrip Order              Ground Beetle Family
##              29              27
##      Rove Beetle Family              Tobacco Aphid
##              27              27
##      Chalcid Wasp              Convergent Lady Beetle
##              25              25
##      Stingless Bee              Spider/Mite Class
##              25              24
##      Tobacco Flea Beetle              Citrus Leafminer
##              24              23
##      Ladybird Beetle              Mason Bee
##              23              22
##      Mosquito              Argentine Ant
##              22              21
```

##	Beetle	Flatheaded Appletree Borer
##	21	20
##	Horned Oak Gall Wasp	Leaf Beetle Family
##	20	20
##	Potato Leafhopper	Tooth-necked Fungus Beetle
##	20	20
##	Codling Moth	Black-spotted Lady Beetle
##	19	18
##	Calico Scale	Fairyfly Parasitoid
##	18	18
##	Lady Beetle	Minute Parasitic Wasps
##	18	18
##	Mirid Bug	Mulberry Pyralid
##	18	18
##	Silkworm	Vedalia Beetle
##	18	18
##	Araneoid Spider Order	Bee Order
##	17	17
##	Egg Parasitoid	Insect Class
##	17	17
##	Moth And Butterfly Order	Oystershell Scale Parasitoid
##	17	17
##	Hemlock Woolly Adelgid Lady Beetle	Hemlock Woolly Adelgid
##	16	16
##	Mite	Onion Thrip
##	16	16
##	Western Flower Thrips	Corn Earworm
##	15	14
##	Green Peach Aphid	House Fly
##	14	14
##	Ox Beetle	Red Scale Parasite
##	14	14
##	Spined Soldier Bug	Armoured Scale Family
##	14	13
##	Diamondback Moth	Eulophid Wasp
##	13	13
##	Monarch Butterfly	Predatory Bug
##	13	13
##	Yellow Fever Mosquito	Braconid Parasitoid
##	13	12
##	Common Thrip	Eastern Subterranean Termite
##	12	12
##	Jassid	Mite Order
##	12	12
##	Pea Aphid	Pond Wolf Spider
##	12	12
##	Spotless Ladybird Beetle	Glasshouse Potato Wasp
##	11	10
##	Lacewing	Southern House Mosquito
##	10	10
##	Two Spotted Lady Beetle	Ant Family
##	10	9
##	Apple Maggot	(Other)
##	9	670

```
#determing 6 most common species in the dataset
summary(Neonics$Species.Common.Name, maxsum=6)
```

```
##           Honey Bee           Parasitic Wasp Buff Tailed Bumblebee
##           667           285           183
## Carniolan Honey Bee           Bumble Bee           (Other)
##           152           140           3196
```

Answer: These species are all polinators, meaning they directly affect agricultural. If the neoin-cotinoids are having adverse affects on these populations, we could see adverse affect in crop yields, and general vegetation health that relies on pollination to reproduce. Additionally, these species live in colonies and their social dynmaics and behaviors could be altered as a colony.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
#looking at the class of one specific column within the dataset
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

Answer: The class is a factor. The column has values like “NR” and numbers, so the whole column cannot be considered numerical.

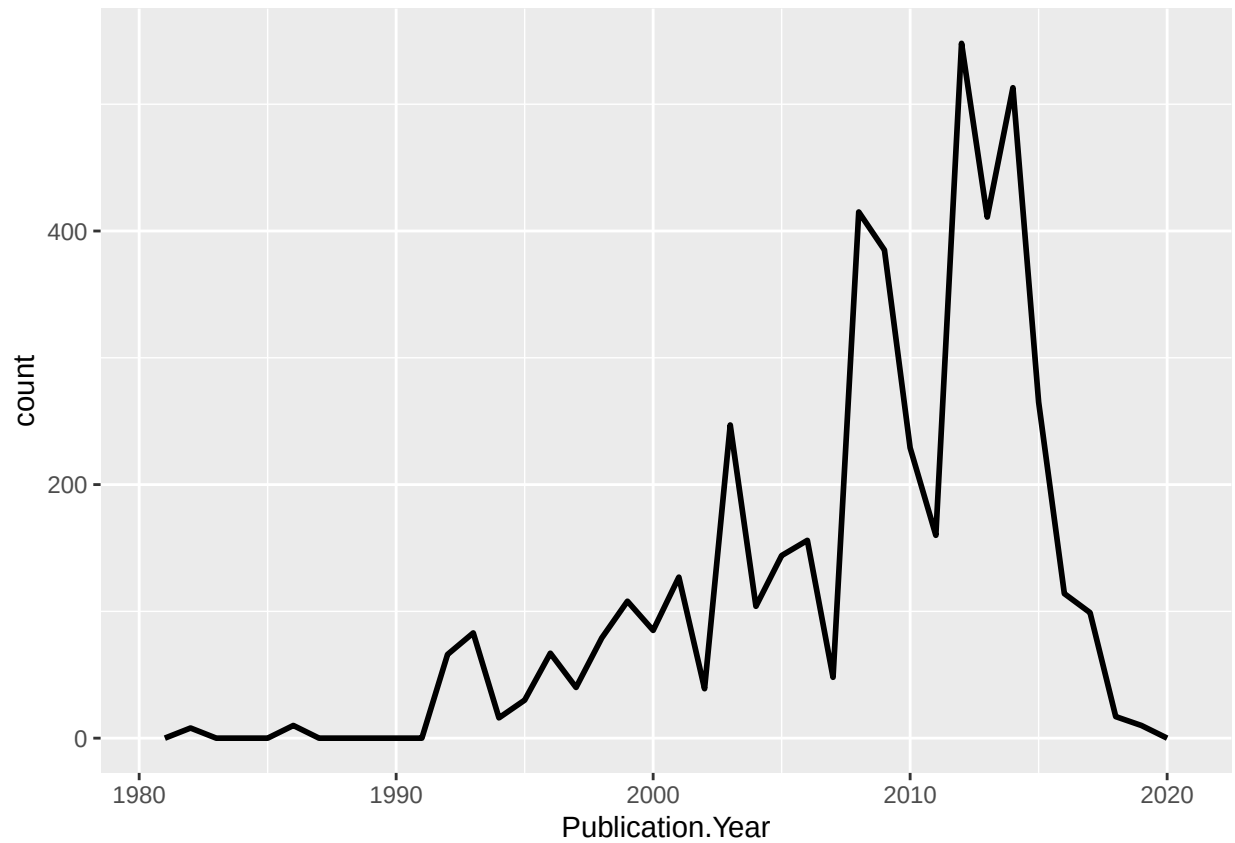
Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

#Plot of studies by publication year, tried different values for binwidth and size, 1 is the most ideal

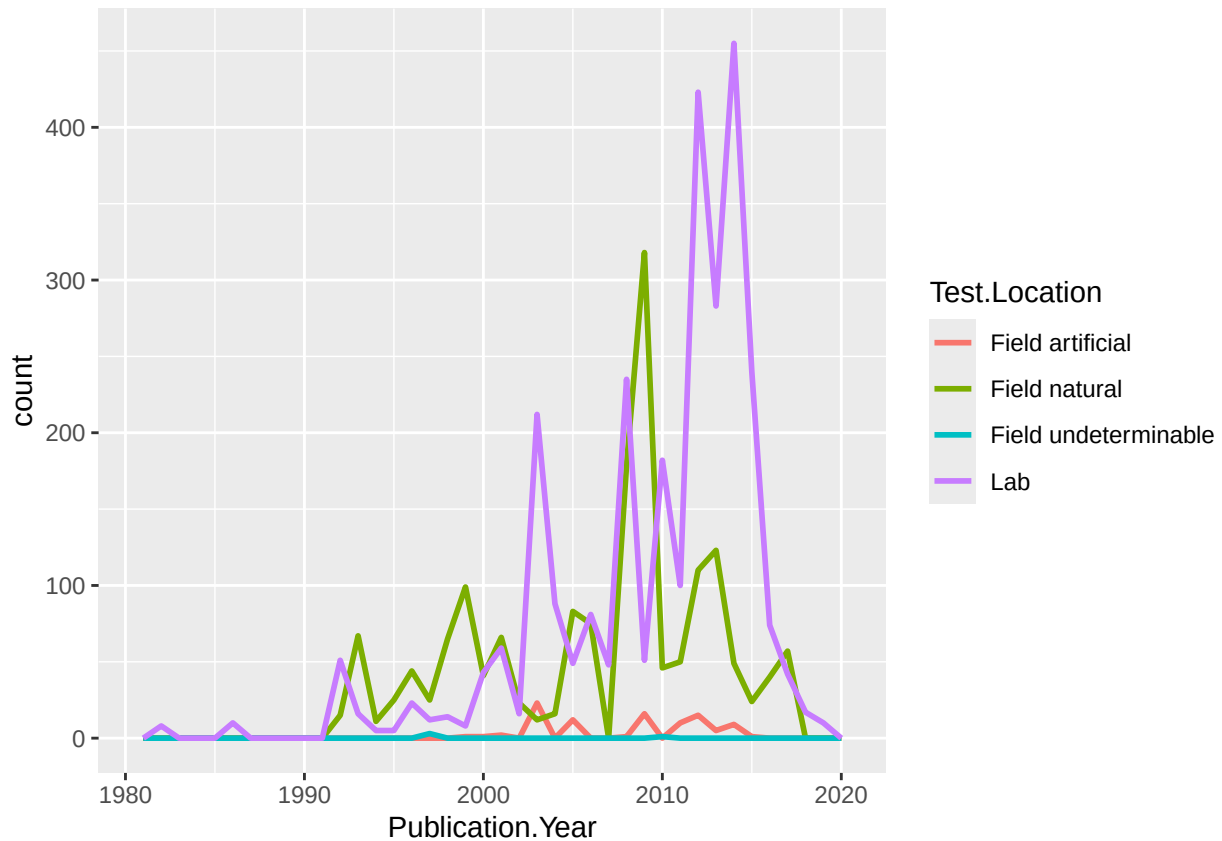
```
ggplot(Neonics, aes(x=Publication.Year))+
  geom_freqpoly(binwidth=1, size=1)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
#adding a color aesthetic so different sites are in different colors  
ggplot(Neonics, aes(x=Publication.Year, color=Test.Location))+  
  geom_freqpoly(binwidth=1, size=1)
```



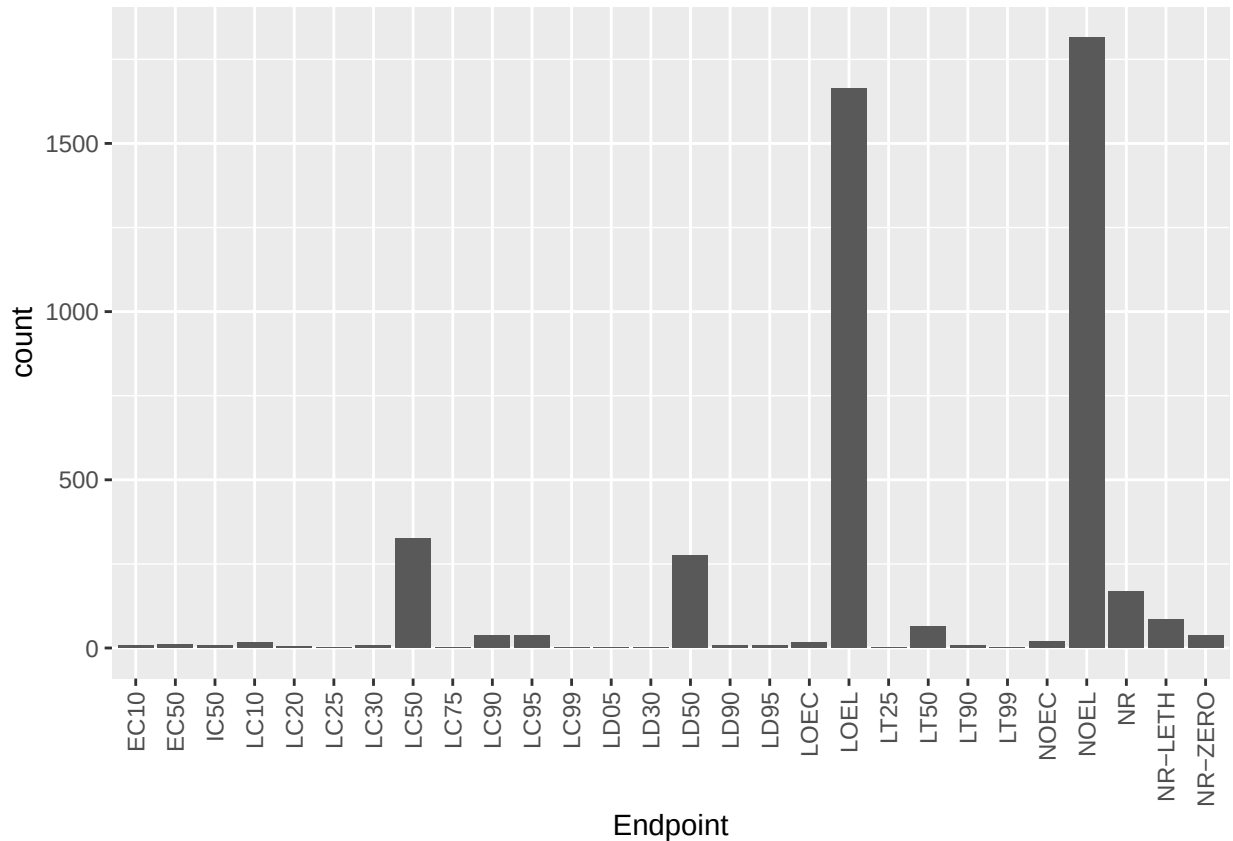
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test locations are in labs. The test locations vary a little over time, with lab sites and field natural sites increasing from the 90s and then field natural sites decrease significantly after 2010.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
#creating barplot and rotating x-axis labels so they're readable
ggplot(Neonics, aes(x=Endpoint))+
  geom_bar()+
  theme(axis.text.x=element_text(angle=90,vjust=0.5,hjust=1))
```

Answer: The two most common endpoints are LOEL (lowest observable effect level) and NOEL (no observable effect level), both in terrestrial context. LOEL means the lowest concentration that produced effects that were significantly different (as reported from responses of controls). NOEL means the opposite, the highest concentration that produced effects not significantly different from responses of controls.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#What class collect date is - it is a factor
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
#converting class from factor to date using lubridate
Litter$collectDate<-ymd(as.character(Litter$collectDate))
#checking new class should be date
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
#Determining which dates litter was sampled in Aug 2018
unique(Litter$collectDate[month(Litter$collectDate)==08&year(Litter$collectDate)==2018])
```

```
## [1] "2018-08-02" "2018-08-30"
```

```
#Data collected on August 2nd and 30th
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
#list of plotIDs
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

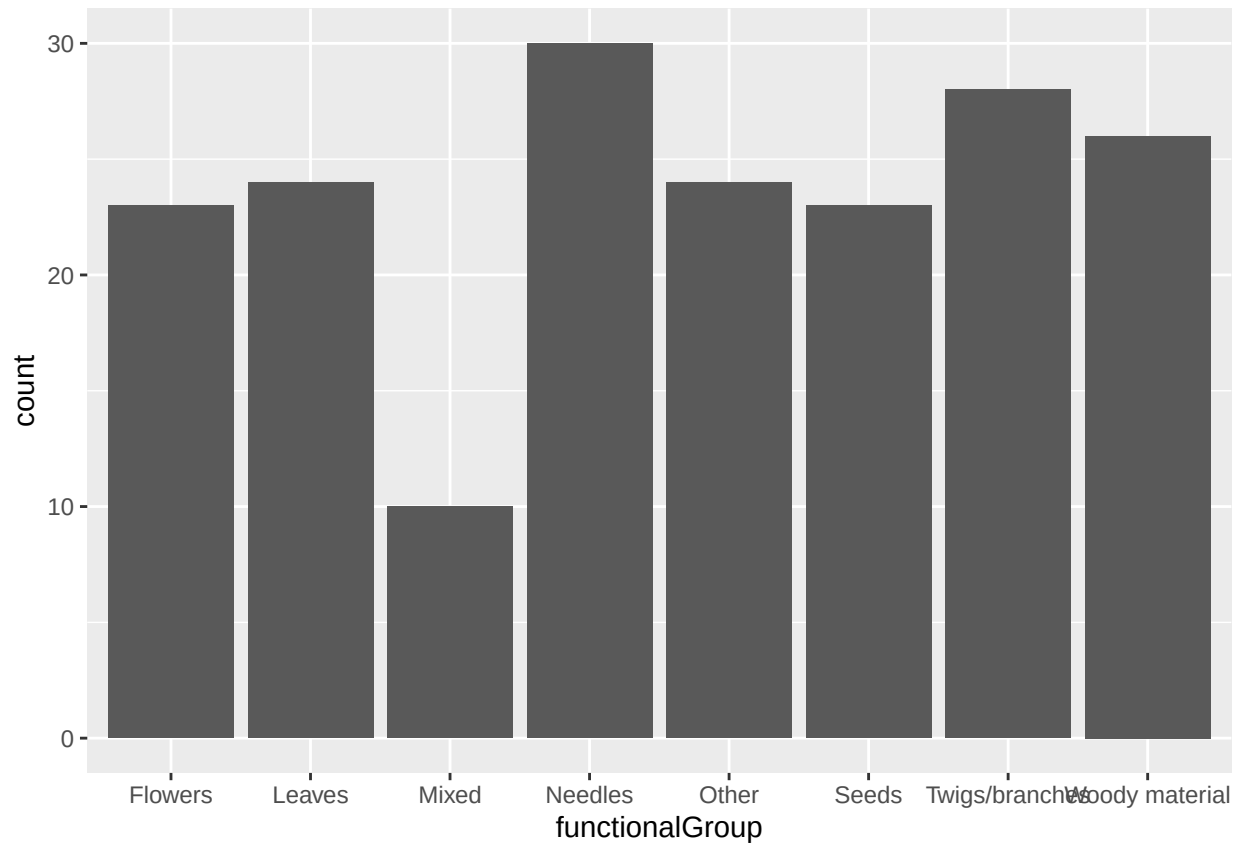
```
#list of plotIDs with counts
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##      20      19      18      15      14      8      16      17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##      14      14      16      17
```

Answer: The `unique` function gives me the plot IDs, and then sorts them by frequency. The `summary` function gives me the plot IDs and their corresponding counts.

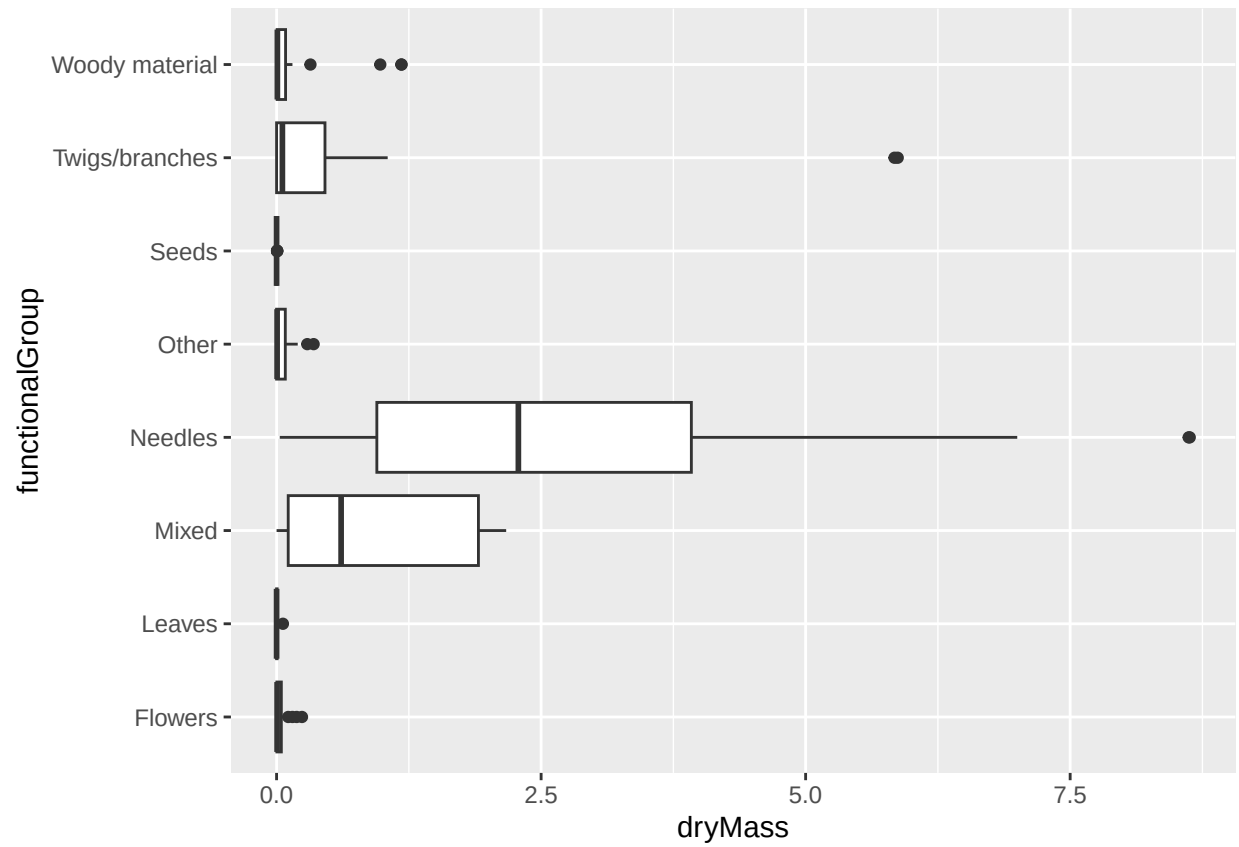
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
#creating bargraph of functional groups
ggplot(Litter, aes(x=functionalGroup))+
  geom_bar()
```

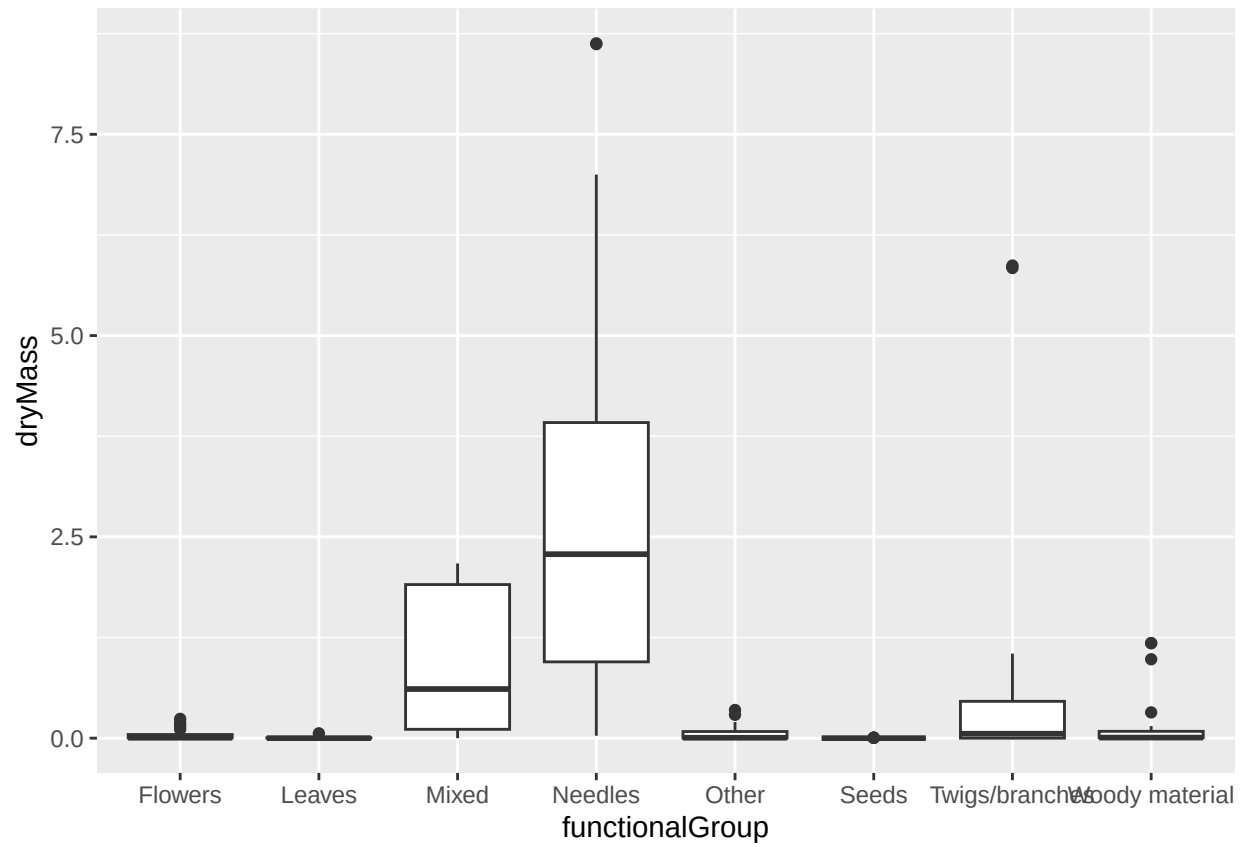


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

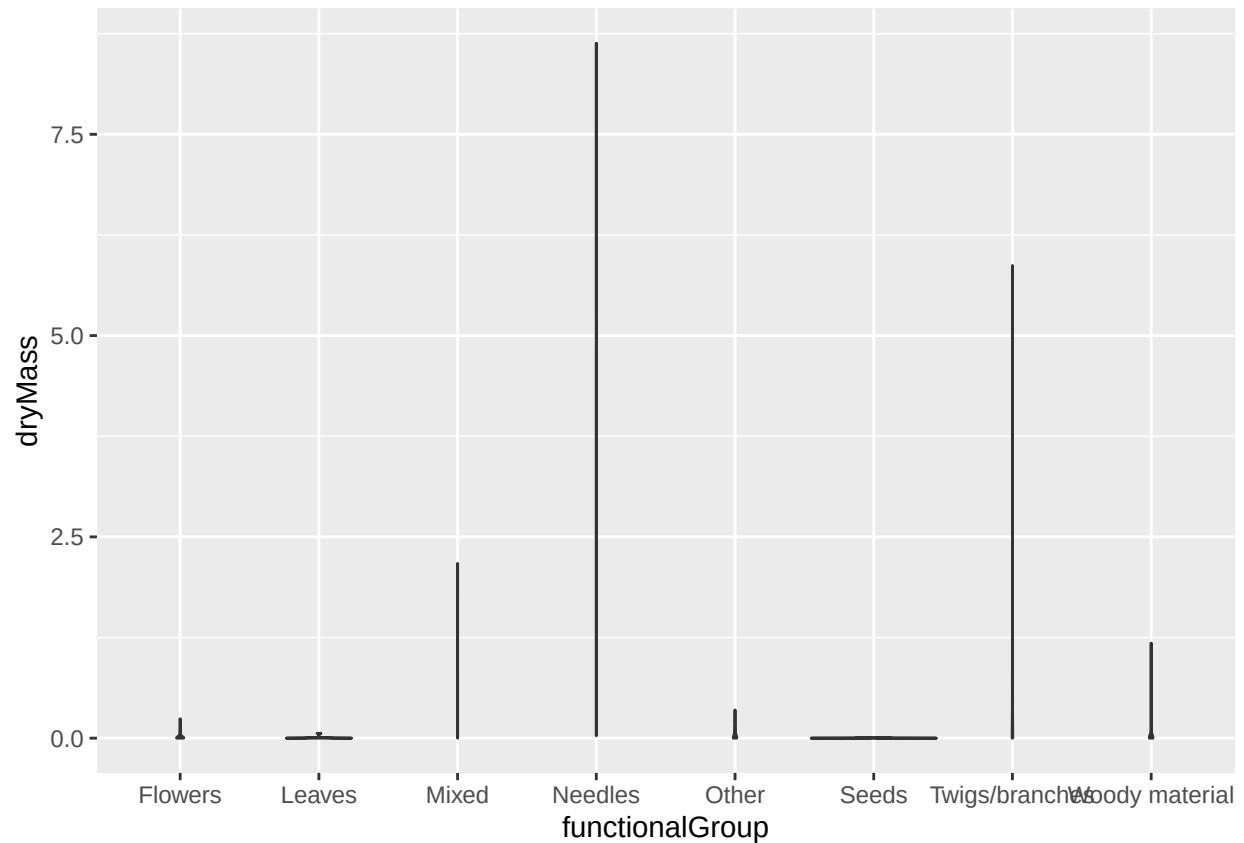
```
#creating boxplot
ggplot(Litter, aes(x=dryMass, y=functionalGroup)) +
  geom_boxplot()
```



```
#first one looked weird - changed x and y
ggplot(Litter,aes(x=functionalGroup, y=dryMass))+
  geom_boxplot()
```



```
#creating violin plot  
ggplot(Litter,aes(x=functionalGroup, y=dryMass))+  
  geom_violin()
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The violin plot reflects density of the observations, and since there isn't much variance in the density there is little to no "violin" shape; we just see vertical lines. The boxplot however more accurately reflects the data as it shows us the median, quartiles and some outliers.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles, mixed litter, and twigs/branches tend to have the highest biomass at these sites.